

Gaussian Belief Propagation for the Diffusion Score

A fully solvable AR(1) chain: factor graph, sum–product, Kalman, and AMP

Giovanni Mantovani

Marco

supervised by M. Mézard and J. Garnier-Brun

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Abstract

We study the *joint* score of a diffusion model on a sequence of correlated frames, in the only setting where everything is closed form: a stationary Gaussian AR(1) Markov chain whose frames are independently corrupted by an Ornstein–Uhlenbeck channel. We derive the clean joint law, the noised joint law $P_t(x) = \mathcal{N}(0, \Sigma_t)$, and the joint score $S(x, t) = -\Sigma_t^{-1}x$, both directly and through Tweedie’s Bayesian (denoiser) identity. We then write the posterior $p(a|x)$ as a factor graph—a *caterpillar tree*—and run belief propagation (BP) on it. We work the $K = 3$ case out fully, message by message, with explicit forward and backward sweeps, and then give the general equations and their Gaussian specialisation, in which every message stays Gaussian by *exact closure* (not a central limit argument): forward filtering is the Kalman filter, the forward–backward combination is the RTS smoother. We verify numerically that BP reproduces the exact score through the precision matrix at machine precision. Finally we contrast BP with approximate message passing (AMP): for this Gaussian model the posterior mean—hence the score—solves the same linear system for mean field, BP, and AMP, so *all three give the same score*; they differ only in the variance estimate, where AMP’s dense-graph closure is accurate at weak coupling but degrades and finally has no solution on the degree-two chain—making precise J. Garnier-Brun’s remark that a central limit theorem “on two points” need not hold. Two experiments close the note: locality (truncated-range messages) versus full sweeps, and the lifecycle of the precision matrix Σ_t^{-1} along diffusion time. All claims are backed by an independent numerical audit (59/59 checks pass); the code and a companion notebook reproduce every figure.

Contents

1	Setup and notation	2
2	The clean joint distribution	2
2.1	Autocovariance and stationarity	2
2.2	Markov factorisation and the tridiagonal precision	3
3	The noisy joint distribution and the score	3
3.1	The score, directly	4
3.2	The score, the Bayesian (Mézar) way	4
4	The posterior and its factor graph	4
4.1	Factor graph	5
5	Belief propagation: the $K = 3$ chain, explicitly	5
5.1	Every message, by hand	6
6	General K and the Gaussian specialisation	7

7	Approximate message passing	8
7.1	The general idea	8
7.2	Our case	8
8	Experiment A: local messages versus full sweeps	9
9	Experiment B: lifecycle of the precision matrix	10
9.1	Shared eigenbasis and the eigenvalue flow	10
9.2	Band-fill lifecycle	10
9.3	One number for the loss of structure	11
10	Summary	11

1 Setup and notation

We follow the notation of Mézard and Montanari [5] and of the handwritten project notebook. A *sequence* is K scalar frames $a = (a_0, a_1, \dots, a_{K-1}) \in \mathbb{R}^K$, indexed by an internal (frame) time $k = 0, \dots, K-1$. The frames are generated by a stationary Gaussian AR(1) Markov chain,

$$a_{k+1} = \alpha a_k + \eta_k, \quad \eta_k \sim \mathcal{N}(0, \sigma_\eta^2) \text{ i.i.d.}, \quad |\alpha| < 1, \quad a_0 \sim \mathcal{N}(0, \sigma_0^2). \quad (1)$$

Throughout we take the *stationary, unit-variance* normalisation

$$\sigma_\eta^2 = 1 - \alpha^2, \quad \sigma_0^2 = \sigma_\infty^2 = \frac{\sigma_\eta^2}{1 - \alpha^2} = 1, \quad (2)$$

so that $\text{Var}(a_k) = 1$ for every k (derived in §2). The OU *diffusion* acts independently on each frame at diffusion time $t \geq 0$:

$$x_k = e^{-t} a_k + \xi_k, \quad \xi_k \sim \mathcal{N}(0, \Delta_t), \quad \Delta_t := 1 - e^{-2t}. \quad (3)$$

This is the variance-preserving OU channel $dx = -x dt + \sqrt{2} dW$: at $t = 0$, $x_k = a_k$; as $t \rightarrow \infty$, $x_k \rightarrow \mathcal{N}(0, 1)$, and indeed $\text{Var}(x_k) = e^{-2t} \cdot 1 + \Delta_t = 1$ for all t .

k	frame index $(0, \dots, K-1)$	t	diffusion (noise) time
a_k	clean frame	x_k	noisy observation of a_k at time t
α	AR(1) coupling	$\sigma_\eta^2 = 1 - \alpha^2$	innovation variance
$\Delta_t = 1 - e^{-2t}$	OU noise variance	Σ_0, Σ_t	clean / noised covariance
$Q_t = \Sigma_t^{-1}$	noised precision (score)	J	posterior precision (§4)
$S(x, t)$	joint score $\nabla_x \log P_t$	$\langle \cdot \rangle_{a x}$	posterior expectation

Remark 1 (Joint, not marginal). The object of interest is the *joint* score $S(x, t) = \nabla_{(x_0, \dots, x_{K-1})} \log P_t(x_0, \dots, x_{K-1}) \in \mathbb{R}^K$, which couples each frame to all the others through the correlations of the chain. The per-frame marginal score $\nabla_{x_k} \log p_t(x_k)$ would treat the frames as independent and is *not* what we study.

2 The clean joint distribution

2.1 Autocovariance and stationarity

Unrolling (1) from a_0 , $a_k = \alpha^k a_0 + \sum_{j=0}^{k-1} \alpha^{k-1-j} \eta_j$.

Derivation : variance of a_k and stationarity

Since $a_0 \perp \eta_j$ and the η_j are i.i.d. with $\text{Cov}(\eta_j, \eta_{j'}) = \sigma_\eta^2 \delta_{jj'}$,

$$\text{Var}(a_k) = \alpha^{2k} \sigma_0^2 + \sigma_\eta^2 \sum_{j=0}^{k-1} \alpha^{2(k-1-j)} = \alpha^{2k} \sigma_0^2 + \sigma_\eta^2 \frac{1 - \alpha^{2k}}{1 - \alpha^2}.$$

With the stationary choice (2), $\sigma_0^2 = \sigma_\eta^2 / (1 - \alpha^2)$, the two terms combine to $\sigma_\eta^2 / (1 - \alpha^2) = 1$ for *all* k : the chain is stationary from frame 0. For $|\alpha| = 1$ the variance grows like $k\sigma_\eta^2$ (random walk); for $|\alpha| > 1$ it grows like α^{2k} (explosive). We keep $|\alpha| < 1$.

By the same computation $\text{Cov}(a_i, a_j) = \alpha^{|i-j|}$, so the clean covariance is the Toeplitz matrix

$$\Sigma_0[i, j] = \alpha^{|i-j|} \quad (\sigma_\infty^2 = 1). \quad (4)$$

2.2 Markov factorisation and the tridiagonal precision

The joint density factorises by the chain rule and the Markov property,

$$P_0(a) = p_0(a_0) \prod_{k=0}^{K-2} M(a_{k+1} | a_k) = \mathcal{N}(a_0; 0, 1) \prod_{k=0}^{K-2} \mathcal{N}(a_{k+1}; \alpha a_k, \sigma_\eta^2). \quad (5)$$

Proposition 1 (Tridiagonal clean precision). $Q_0 := \Sigma_0^{-1}$ is tridiagonal:

$$(Q_0)_{kk} = \frac{1 + \alpha^2}{\sigma_\eta^2} \text{ (interior)}, \quad (Q_0)_{00} = (Q_0)_{K-1, K-1} = \frac{1}{\sigma_\eta^2}, \quad (Q_0)_{k, k\pm 1} = -\frac{\alpha}{\sigma_\eta^2}.$$

Derivation 1: from the log-density

$-2 \log P_0(a) = a_0^2 + \sum_k (a_{k+1} - \alpha a_k)^2 / \sigma_\eta^2 + \text{const.}$ The coefficient of a_k^2 collects one term from each adjacent transition, giving $(1 + \alpha^2) / \sigma_\eta^2$ in the interior; the coefficient of $a_k a_{k+1}$ is $-\alpha / \sigma_\eta^2$; there is no $a_k a_{k+d}$ term for $d \geq 2$. Hence $Q_0 = \nabla^2(-\log P_0)$ is tridiagonal.

Reading : tridiagonality is Markovianity, not Gaussianity

Σ_0 (covariance) is *dense*; its inverse Q_0 (precision) is *tridiagonal*. A zero in the precision means conditional independence given the rest. The chain couples only neighbours, so all entries beyond the first off-diagonal vanish. This sparsity is the signature of the *Markov* structure, and it is the only sparse object in the whole story.

3 The noisy joint distribution and the score

The channel (3) is applied independently to each frame, so given the clean sequence, $p_t(x | a) = \prod_{k=0}^{K-1} \mathcal{N}(x_k; e^{-t} a_k, \Delta_t)$, and the noised joint law marginalises over the clean sequence:

$$P_t(x) = \int_{\mathbb{R}^K} P_0(a) \prod_{k=0}^{K-1} \mathcal{N}(x_k; e^{-t} a_k, \Delta_t) da. \quad (6)$$

Proposition 2 (Noised covariance). P_t is the zero-mean Gaussian $P_t(x) = \mathcal{N}(x; 0, \Sigma_t)$ with

$$\Sigma_t = e^{-2t} \Sigma_0 + \Delta_t I_K. \quad (7)$$

Derivation 2: mean and covariance of x

$\mathbb{E}[x_k] = e^{-t}\mathbb{E}[a_k] = 0$. For the covariance, with $x_i = e^{-t}a_i + \xi_i$ and $\xi \perp a$,

$$\text{Cov}(x_i, x_j) = e^{-2t} \text{Cov}(a_i, a_j) + \mathbb{E}[\xi_i \xi_j] = e^{-2t}(\Sigma_0)_{ij} + \Delta_t \delta_{ij}.$$

A linear-Gaussian marginalisation of a Gaussian is Gaussian, so (7) holds.

3.1 The score, directly

Since $\log P_t(x) = -\frac{1}{2}x^\top \Sigma_t^{-1}x + \text{const}$ and $\nabla_x(x^\top Ax) = (A + A^\top)x = 2\Sigma_t^{-1}x$ for symmetric $A = \Sigma_t^{-1}$,

$$S(x, t) = \nabla_x \log P_t(x) = -\Sigma_t^{-1}x = -Q_t x \quad Q_t := \Sigma_t^{-1}. \quad (8)$$

The score is *linear* in x (Gaussianity), and its coupling structure is exactly that of the precision Q_t .

3.2 The score, the Bayesian (Mézard) way

Bayesian view : the score is a denoiser (Tweedie)

Differentiating (6) under the integral and using $\partial_{x_k} \mathcal{N}(x_k; e^{-t}a_k, \Delta_t) = \frac{e^{-t}a_k - x_k}{\Delta_t} \mathcal{N}(\dots)$,

$$\partial_{x_k} \log P_t(x) = \frac{1}{P_t(x)} \int \frac{e^{-t}a_k - x_k}{\Delta_t} P_0(a) p_t(x|a) da = \left\langle \frac{e^{-t}a_k - x_k}{\Delta_t} \right\rangle_{a|x},$$

because $P_0(a)p_t(x|a) = P_t(x)p(a|x)$ (Bayes). Hence the *Tweedie identity* [1, 7]

$$S_k(x, t) = \frac{e^{-t} \langle a_k \rangle_{a|x} - x_k}{\Delta_t} \quad (9)$$

where $\langle a_k \rangle_{a|x}$ is the posterior mean of the clean frame. To compute the k -th component of the joint score we must know the posterior mean of a_k given the *whole* noisy sequence x .

Reading : the two forms agree

Equations (8) and (9) are the same vector. Indeed the posterior mean is linear, $\langle a \rangle_{a|x} = C_t x$ for a matrix C_t (§4), and substituting into (9) returns $-Q_t x$. The numerical audit confirms $\|S_{\text{matrix}} - S_{\text{Tweedie}}\| < 10^{-12}$.

4 The posterior and its factor graph

By Bayes, the posterior over the clean sequence given the (fixed) observations x is

$$p(a|x) \propto P_0(a) p_t(x|a) = \underbrace{\mathcal{N}(a_0; 0, 1)}_{\psi_{\text{pr}}(a_0)} \prod_{k=0}^{K-2} \underbrace{\mathcal{N}(a_{k+1}; \alpha a_k, \sigma_\eta^2)}_{\psi_{\text{tr}}^{(k)}(a_k, a_{k+1})} \prod_{k=0}^{K-1} \underbrace{\mathcal{N}(x_k; e^{-t}a_k, \Delta_t)}_{\psi_{\text{ob}}^{(k)}(a_k)}. \quad (10)$$

It is a product of Gaussian factors, hence itself Gaussian. Completing the square gives

$$p(a|x) = \mathcal{N}(a; m, J^{-1}), \quad J = \frac{e^{-2t}}{\Delta_t} I + Q_0, \quad m = J^{-1}h, \quad h = \frac{e^{-t}}{\Delta_t} x. \quad (11)$$

The posterior precision J is again tridiagonal (Q_0 tridiagonal plus a diagonal), and $\langle a \rangle_{a|x} = m = J^{-1}h$ is linear in x , as promised.

4.1 Factor graph

A *factor graph* is the bipartite graph whose variable nodes are the a_k and whose factor nodes are the local factors of (10); an edge joins a factor to a variable iff the variable is an argument of the factor [4]. The observations x_k are *not* variable nodes—they are clamped constants inside the observation factors $\psi_{\text{ob}}^{(k)}$. For K frames there are $2K$ factors (1 prior + $(K-1)$ transitions + K observations).

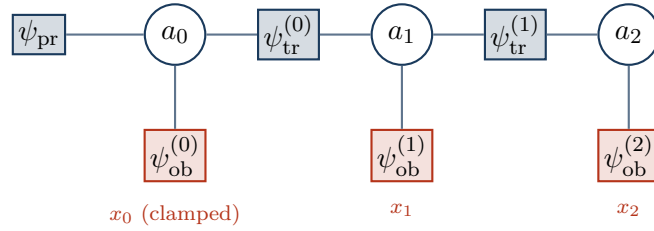


Figure 1: Factor graph of $p(a|x)$ for $K=3$. Circles: variable (clean frame) nodes. Blue squares: prior and transition factors. Red squares: observation factors carrying the clamped data x_k . The graph is a *caterpillar tree*—a backbone $a_0-a_1-a_2$ with one leaf factor per node—hence has no cycles.

Proposition 3 (Tree $\Rightarrow$ BP exact). *The factor graph in Fig. 1 is connected and acyclic. By the tree-exactness of belief propagation [5, Th. 14.1], the BP marginals equal the exact marginals of $p(a|x)$.*

Reading : this is exactly the geometry J. Garnier-Brun described

The graph is “very tree-like—a straight line with things on the side.” Every interior frame node has degree three (two transition factors + one observation), and every BP update combines only a *small, fixed* number of incoming messages. This is the opposite of the dense graphs for which AMP is designed (§7).

5 Belief propagation: the $K=3$ chain, explicitly

We first recall the general sum-product rules [4, 5], then execute them by hand on Fig. 1. There are two kinds of message, both functions of the receiving variable: variable-to-factor $\nu_{a_k \rightarrow \psi}$ and factor-to-variable $\hat{\nu}_{\psi \rightarrow a_k}$,

$$\nu_{i \rightarrow a}(a_i) \propto \prod_{b \in \partial i \setminus a} \hat{\nu}_{b \rightarrow i}(a_i), \quad (12)$$

$$\hat{\nu}_{a \rightarrow i}(a_i) \propto \int \psi_a(a_{\partial a}) \prod_{j \in \partial a \setminus i} \nu_{j \rightarrow a}(a_j) da_{\partial a \setminus i}, \quad (13)$$

$$p(a_i) \propto \prod_{a \in \partial i} \hat{\nu}_{a \rightarrow i}(a_i). \quad (14)$$

On a chain these collapse into a single *forward* sweep and a single *backward* sweep. We use *Convention A*: the forward message into a_k carries the evidence $x_0, \dots, x_{k-1}$ only, the backward message carries $x_{k+1}, \dots, x_{K-1}$ only, and the local evidence $\psi_{\text{ob}}^{(k)}$ is multiplied in at combination time—never inside a recursion. This is the one bookkeeping that avoids double-counting x_k .

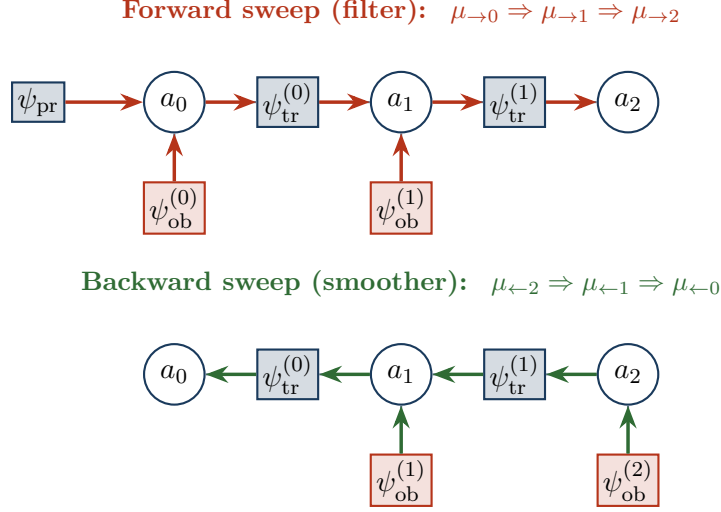


Figure 2: The two sweeps for $K = 3$. Forward (red): information from the prior and from x_0, x_1 flows left to right. Backward (green): information from x_2, x_1 flows right to left. The posterior at a_k multiplies the forward message, the local evidence $\psi_{\text{ob}}^{(k)}$, and the backward message.

5.1 Every message, by hand

Write each Gaussian message in *information form* $\mu(a) \propto \exp(-\frac{1}{2}\lambda a^2 + \theta a)$, i.e. mean θ/λ and variance $1/\lambda$. The local observation factor, read as a function of a_k , is

$$\psi_{\text{ob}}^{(k)}(a_k) = \mathcal{N}(x_k; e^{-t}a_k, \Delta_t) \propto \exp\left(-\frac{1}{2}\frac{e^{-2t}}{\Delta_t}a_k^2 + \frac{e^{-t}}{\Delta_t}x_k a_k\right) \Leftrightarrow (\lambda_{\text{ob}}, \theta_{\text{ob}}) = \left(\frac{e^{-2t}}{\Delta_t}, \frac{e^{-t}}{\Delta_t}x_k\right). \quad (15)$$

Derivation : forward sweep for $K = 3$ (Convention A)

The forward message starts at the prior, $\mu_{\rightarrow 0} = \psi_{\text{pr}} = \mathcal{N}(a_0; 0, 1)$, i.e. $(\lambda, \theta) = (1, 0)$. At each node we combine the forward message with the local evidence (15) (precisions and potentials add),

$$\lambda_k^c = \lambda_k^{\rightarrow} + \lambda_{\text{ob}}, \quad \theta_k^c = \theta_k^{\rightarrow} + \frac{e^{-t}}{\Delta_t}x_k, \quad m_k^c = \theta_k^c / \lambda_k^c, \quad v_k^c = 1 / \lambda_k^c,$$

and push the result through the transition $\psi_{\text{tr}}^{(k)}$ to the next node. The predict step uses $\int \mathcal{N}(a_{k+1}; \alpha a_k, \sigma_\eta^2) \mathcal{N}(a_k; m_k^c, v_k^c) da_k = \mathcal{N}(a_{k+1}; \alpha m_k^c, \alpha^2 v_k^c + \sigma_\eta^2)$, so

$$\mu_{\rightarrow k+1} = \mathcal{N}(a_{k+1}; \alpha m_k^c, \alpha^2 v_k^c + \sigma_\eta^2).$$

Concretely for $K = 3$: $\mu_{\rightarrow 0} = (1, 0) \rightarrow$ combine $x_0 \rightarrow$ predict $\Rightarrow \mu_{\rightarrow 1} \rightarrow$ combine $x_1 \rightarrow$ predict $\Rightarrow \mu_{\rightarrow 2}$. This is exactly the *Kalman filter* predict–update cycle.

Derivation : backward sweep for $K = 3$

The backward message starts flat at the last node, $\mu_{\leftarrow 2} \equiv 1$ (precision $\lambda = 0$, uninformative). At node k we combine $\mu_{\leftarrow k}$ with the local evidence to get (m_k^c, v_k^c) and back-propagate through the transition $a_k = \alpha a_{k-1} + \eta_{k-1}$:

$$\mu_{\leftarrow k-1} = \mathcal{N}\left(a_{k-1}; \frac{m_k^c}{\alpha}, \frac{v_k^c + \sigma_\eta^2}{\alpha^2}\right).$$

For $K = 3$: $\mu_{\leftarrow 2} \equiv 1 \rightarrow$ combine $x_2 \rightarrow$ back-propagate $\Rightarrow \mu_{\leftarrow 1} \rightarrow$ combine $x_1 \rightarrow$ back-propagate $\Rightarrow \mu_{\leftarrow 0}$.

Derivation : combination $\Rightarrow$ posterior marginal

At every node the exact posterior marginal multiplies the three pieces meeting there—forward message, local evidence, backward message,

$$p(a_k | x) \propto \mu_{\rightarrow k}(a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k),$$

i.e. in information form the precisions and potentials of the three add. The posterior mean is $\langle a_k \rangle_{a|x} = \theta_k^{\text{post}} / \lambda_k^{\text{post}}$ and the variance is $1/\lambda_k^{\text{post}}$. Plugging the mean into the Tweedie identity (9) returns the score. This forward-backward combination is the *Rauch-Tung-Striebel smoother* [3, 6].

The companion notebook prints the numeric (mean, variance) of every message for a concrete $K = 3$ instance and checks the recombination against the matrix posterior (11).

6 General K and the Gaussian specialisation

For general K the same two recursions hold. Writing the forward (*filtered*) message into a_k as $\mu_{\rightarrow k}$ and the backward as $\mu_{\leftarrow k}$, the sum-product rules (12)–(13) on the chain are

$$\mu_{\rightarrow k+1}(a_{k+1}) = \int M(a_{k+1} | a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\rightarrow k}(a_k) da_k, \quad \mu_{\rightarrow 0} = p_0, \quad (16)$$

$$\mu_{\leftarrow k-1}(a_{k-1}) = \int M(a_k | a_{k-1}) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) da_k, \quad \mu_{\leftarrow K-1} \equiv 1, \quad (17)$$

and the marginal is the product $p(a_k | x) \propto \mu_{\rightarrow k} \psi_{\text{ob}}^{(k)} \mu_{\leftarrow k}$. Each sweep is $K - 1$ one-dimensional integrals, so BP costs $O(K)$ versus the $O(K^3)$ of forming J^{-1} directly.

Reading : why the Gaussian case is special—answering the doubt about “are the messages Gaussian?”

Each operation in (16)–(17) maps Gaussians to Gaussians: the *product* of two Gaussians is Gaussian (precisions add), and the *integral* of a Gaussian against a Gaussian transition kernel is Gaussian (predict step). Therefore, started from Gaussian factors, every message is *exactly* Gaussian and is carried by just two numbers (λ, θ) . This closure is *algebraic*—it holds for any node degree, on the two-input chain just as well as on a dense graph. It is *not* a central limit approximation. Consequently the posterior is exactly Gaussian and BP is exact: the guess that “BP in the Gaussian case recovers the right solution” is correct, and the reason is closure, not averaging.

Numerical check : BP reproduces the exact score

`numerical_audit.py` runs `bp_score.py` against the matrix posterior (11) and the closed-form score (8) over 80 configurations ($K \in \{2, 3, 5, 8, 16\}$, $\alpha \in \{0.2, 0.5, 0.9, -0.4\}$, $t \in \{0.05, 0.3, 1, 3\}$). Maximum errors: posterior mean 1.3×10^{-15} , posterior variance 3.9×10^{-15} , joint score 1.2×10^{-14} . BP *is* the score, recovered through the precision matrix.

7 Approximate message passing

7.1 The general idea

AMP (in physics, the TAP equations [8]) is the message passing one uses when BP messages cannot be represented exactly and the graph is *dense*. For a linear model $y = \Phi a + w$ with a dense measurement matrix $\Phi \in \mathbb{R}^{m \times N}$ and a separable prior, the factor graph is the complete bipartite graph: every factor touches *all* N variables (Fig. 1 of Genovese and Piana [2]). Each message is then a sum of many weakly dependent contributions, so by a central limit argument it is *approximately Gaussian*, and one propagates only its mean and variance. The defining feature of AMP relative to naive iteration is the *Onsager reaction term*, which subtracts each node’s own back-influence; the variances become label-independent and the means obey a closed recursion—the AMP equations. Rigorously, Genovese and Piana [2] show (for ℓ_2 minimisation) that the BP means satisfy the AMP equations *asymptotically* as $N \rightarrow \infty$, and note that on a *tree* BP already reconstructs the measure exactly.

7.2 Our case

Our model is the opposite of the AMP regime: the measurement operator is *diagonal* ($x = e^{-t}a + \text{noise}$), the coupling lives in the *prior* (the AR(1) chain), and the factor graph is a degree- ≤ 2 tree. None of AMP’s structural assumptions (dense Φ , many neighbours, separable prior) hold. To compare BP and AMP concretely we view the Gaussian posterior (11) as a Gaussian Markov random field with the tridiagonal precision J and field h , and ask three solvers for the marginals.

Reading : the mean is shared, the variance is not

The posterior *mean* solves the linear system $Jm = h$. Mean field, BP, and AMP are all iterative solvers of that *same* system, so at their fixed point they return the *same exact mean*—hence, via Tweedie (9), the *same score*. They differ only in the per-node *variance* closure:

$$\text{mean field: } V_i = \frac{1}{J_{ii}}; \quad \text{AMP/TAP: } V_i = \frac{1}{J_{ii} - \sum_{k \neq i} J_{ik}^2 V_k}; \quad \text{BP/exact: } V_i = (J^{-1})_{ii}.$$

The AMP closure is the cavity variance with the “exclude one neighbour” correction dropped ($V_{k \rightarrow i} \approx V_k$). Dropping it is harmless when each node has *many* weak neighbours—removing one changes the cavity by $O(1/N)$, the dense regime where AMP is exact—but on the chain each node has *two* neighbours and removing one is an $O(1)$ change. This is precisely J. Garnier-Brun’s observation that “a central limit theorem on two data points is usually not right.”

Numerical check : BP versus AMP, measured

On a *dense* random SPD precision ($N = 600$) the AMP variance matches the exact $\text{diag}(J^{-1})$ to 1.7×10^{-3} , far better than mean field (1.3×10^{-1}): AMP works where it is designed to. On the *chain* (Fig. 3, $K = 9$, $\alpha = 0.8$): the AMP mean equals the exact mean to $\sim 10^{-12}$ for every t (**BP and AMP give the same score**); the AMP variance is accurate at small t (evidence-dominated, weak coupling, error 1.3×10^{-4} at $t = 0.05$) but *has no physical fixed point* once the coupling is strong enough (here $t \gtrsim 0.4$): the self-consistent denominator $J_{ii} - \sum_k J_{ik}^2 V_k$ turns negative. BP stays exact throughout.

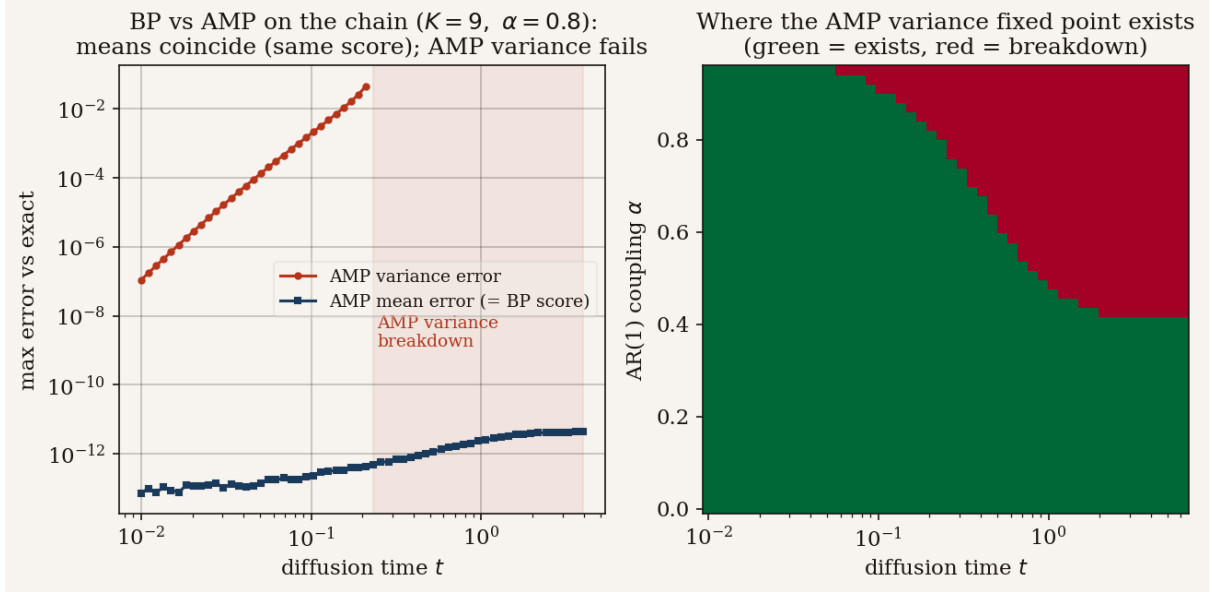


Figure 3: **BP vs. AMP on the chain.** Left: the AMP *mean* error (blue, $\approx 10^{-12}$) is flat in t —the score is closure-independent, so BP and AMP coincide—while the AMP *variance* error (red) grows and then the closure breaks down (shaded). Right: existence of the AMP variance fixed point over (α, t) ; green = exists (weak coupling / small t), red = breakdown (strong coupling). BP is exact everywhere.

8 Experiment A: local messages versus full sweeps

How much does a frame’s estimate depend on *distant* frames? Define the *local estimator of radius r* as the posterior mean of a_k given only the observations within distance r , $\langle a_k \rangle_a | x_{k-r:k+r}$, the rest of the chain marginalised out. Because any contiguous block of a stationary AR(1) chain is itself a stationary AR(1) chain, this is computed exactly on the windowed sub-chain—the only approximation is the truncation of range. Taking $r \geq K - 1$ recovers full BP, which is exact; so the gap $\|S^{\text{local},r} - S^{\text{full}}\|$ isolates the error due to locality.

Reading : in the Gaussian case the local approximation is controlled

Full BP is exact, so locality is the *only* error, and it is governed by the decay of correlations ($\sim \alpha^{\text{distance}}$) and by how far the precision Q_t reaches off the diagonal. A strictly-neighbour scheme ($r = 1$) is exact at $t = 0$ (where $Q_t = Q_0$ is tridiagonal) and accurate again at large t , but misses the filled band at intermediate t . The Gaussian model thus makes the local/discrete approximation *quantitative*.

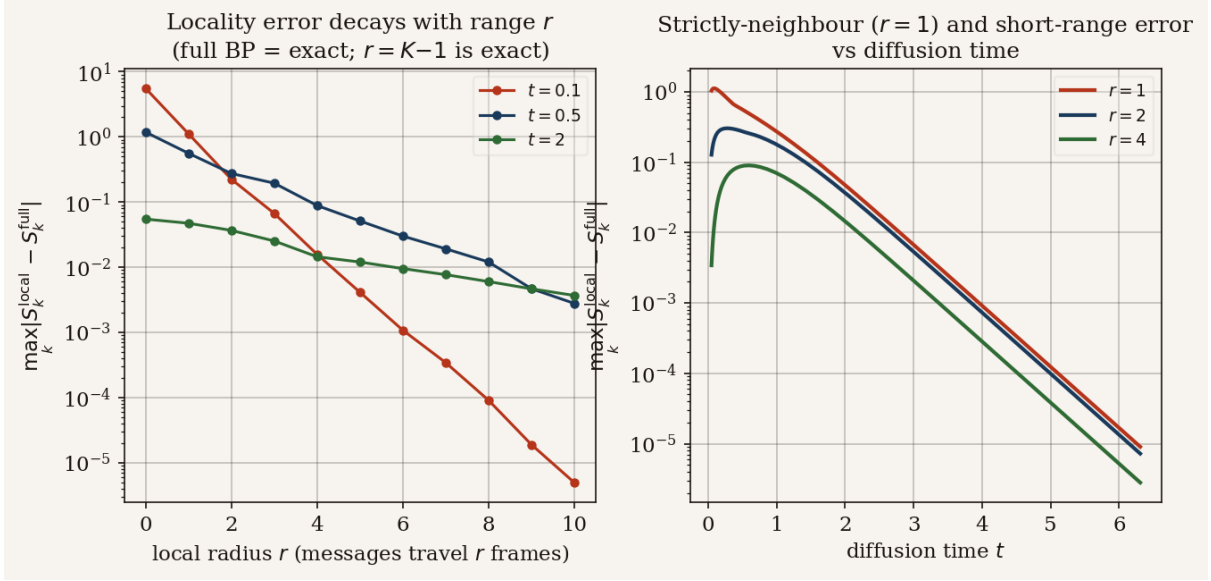


Figure 4: **Locality.** Left: the error decays geometrically with the range r and vanishes at $r = K - 1$ (full chain, exact). Right: at fixed range the error is largest at *intermediate* t —small at $t \rightarrow 0$ (each frame is pinned by its own observation) and at $t \rightarrow \infty$ (the score decouples, $S \rightarrow -x$). The decay rate is set by the off-diagonal reach of Q_t (§9): the more the band has filled, the longer the range needed.

9 Experiment B: lifecycle of the precision matrix

The score (8) is entirely controlled by $Q_t = \Sigma_t^{-1}$. Its sparsity is the conditional-independence structure of P_t as a Gaussian MRF (a zero in Q_t = a conditional independence). We track how that structure evolves with diffusion time.

9.1 Shared eigenbasis and the eigenvalue flow

Because $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$ adds only a multiple of the identity, the eigenvectors of Σ_t and of Q_t are exactly those of Σ_0 for every t ; only the eigenvalues move,

$$\Sigma_0 = U \text{diag}(\lambda_i) U^\top \Rightarrow \lambda_i(\Sigma_t) = e^{-2t}\lambda_i + \Delta_t \rightarrow 1, \quad \lambda_i(Q_t) = \frac{1}{e^{-2t}\lambda_i + \Delta_t} \rightarrow 1. \quad (18)$$

Since Σ_0 is symmetric Toeplitz, its eigenvectors are near-Fourier (cosine) modes (Fig. 5); diffusion contracts every eigenvalue toward 1, so $Q_t \rightarrow I$ and the score $\rightarrow -x$.

9.2 Band-fill lifecycle

In the *frame* basis the story is different and is the one of direct interest: Q_t is tridiagonal only at $t = 0$; for $t > 0$ the band fills, and at large t it collapses back to a (scaled) identity. Figure 6 shows the three regimes.

The filling is quantitative. From the resolvent $Q_t = e^{2t} \sum_{m \geq 0} (-c_t)^m Q_0^{m+1}$ with $c_t = e^{2t} - 1 = 2t + O(t^2)$, the leading distance- d entry comes from $m = d - 1$:

$$(Q_t)_{i,i+d} = (-1)^{d-1} (2t)^{d-1} (Q_0^d)_{i,i+d} + O(t^d). \quad (19)$$

Remark 2 (Correction to a published coefficient). An earlier internal note (`precision_lifecycle_summary.pdf`, eq. 12) printed (19) with a spurious $1/(d-1)!$ factor. That factor does *not* follow from the resolvent expansion: the leading distance- d term is

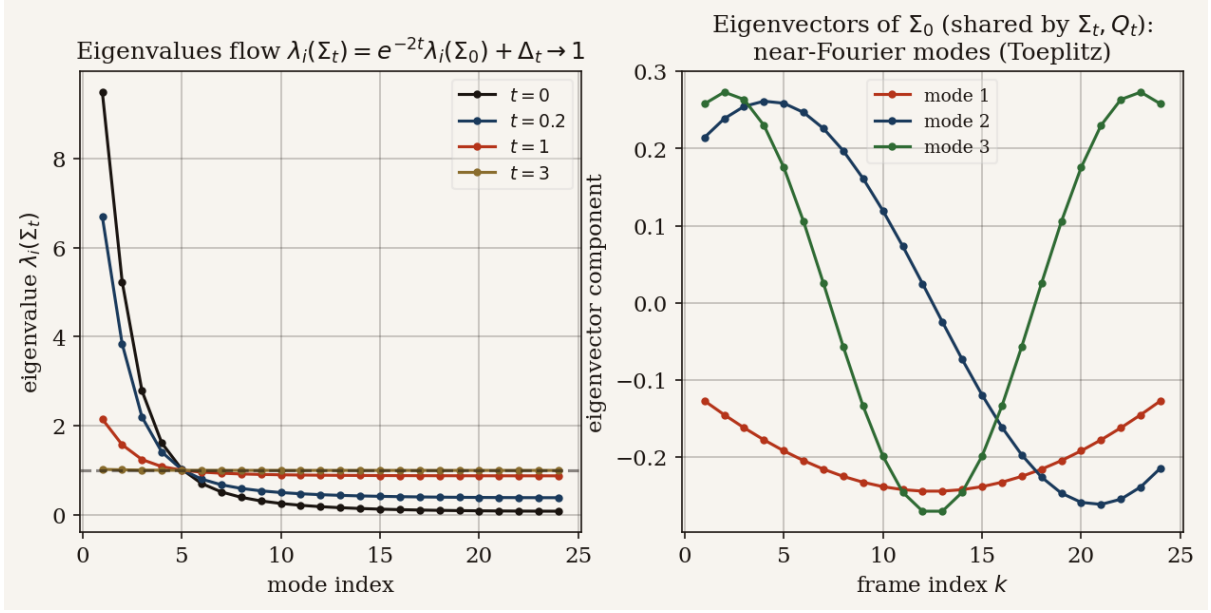


Figure 5: **Spectral decomposition.** Left: the eigenvalues $\lambda_i(\Sigma_t) = e^{-2t}\lambda_i(\Sigma_0) + \Delta_t$ all flow to 1. Right: the shared eigenvectors of Σ_0 (hence of Σ_t, Q_t) are near-Fourier modes. In this basis the multivariate denoising decouples into K scalar problems.

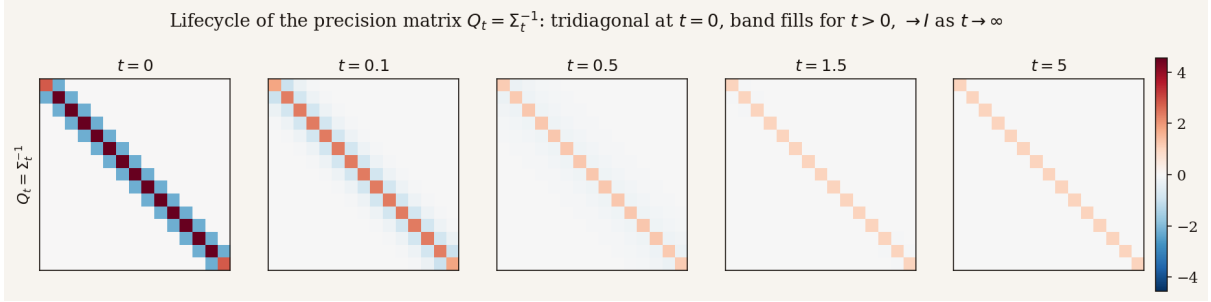


Figure 6: **Lifecycle of $Q_t = \Sigma_t^{-1}$.** Tridiagonal at $t = 0$ (Markov); the band fills for $t > 0$ (dense); the magnitude collapses and $Q_t \rightarrow I$ as $t \rightarrow \infty$ (isotropic Gaussian).

the single $m = d - 1$ contribution $(-c_t)^{d-1}$, with no combinatorial denominator. The numerical audit confirms the version above (no factorial) to better than 1% for $d = 1, \dots, 5$; the factorial version fails the same test by factors that reach 800% at $d = 5$.

9.3 One number for the loss of structure

To summarise the lifecycle with a single curve, Fig. 8 plots $\|\Sigma_t - \Sigma_0\|_F$ (how far the covariance has moved) and the off-tridiagonal Frobenius mass of Q_t (how non-tridiagonal the precision is). The latter rises from zero, peaks at intermediate t , and falls back to zero as $Q_t \rightarrow I$ —the band fills and then empties.

10 Summary

- The joint score of the Gaussian AR(1)+OU model is linear, $S(x, t) = -\Sigma_t^{-1}x$, equal to the Tweedie denoiser $S_k = (e^{-t}\langle a_k \rangle_{a|x} - x_k)/\Delta_t$.
- The posterior is a Gaussian caterpillar tree; BP is exact. The forward sweep is the Kalman filter, the forward-backward combination the RTS smoother, and the messages are Gaussian

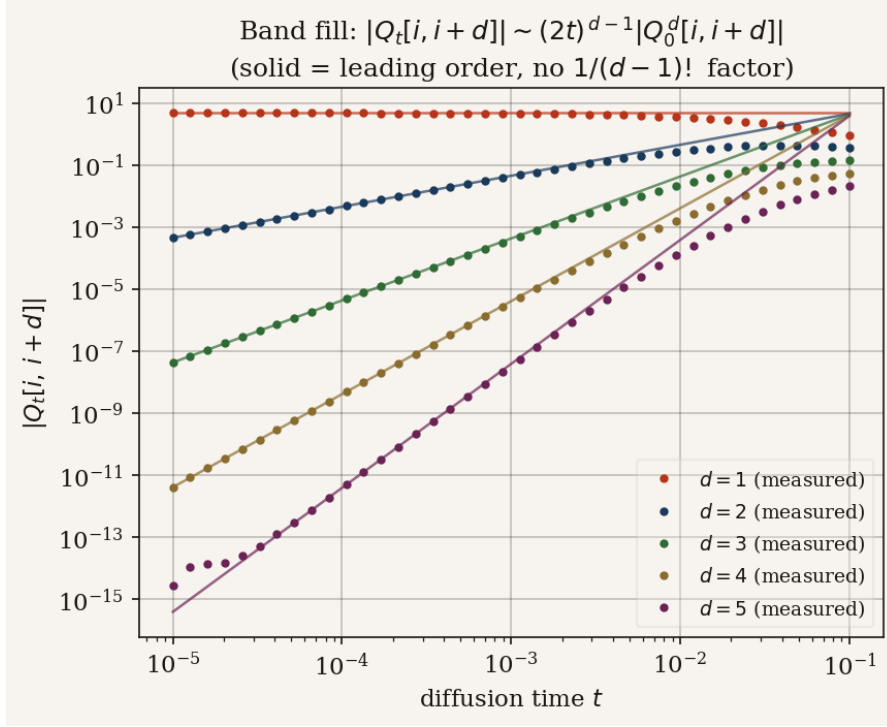


Figure 7: **Band fill, log-log.** Measured $|Q_t[i, i+d]|$ (dots) against the leading order (19) (solid, no factorial). Slopes $d-1 = 0, 1, 2, 3, 4$; agreement is exact at small t , with the expected departure as t grows and the noise floor at $d = 5$.

by *exact closure*—resolving the doubt about Gaussianity without invoking any central limit theorem.

- BP reconstructs Σ_t^{-1} and hence the score at machine precision.
- BP and AMP give the *same score* (the mean solves the same linear system); they differ only in the variance, where AMP’s dense-graph closure is accurate at weak coupling but breaks down on the degree-two chain—making precise why “a CLT on two points” need not hold.
- Locality (Exp. A) and the band-fill lifecycle of Q_t (Exp. B) are two views of the same off-diagonal decay: the score is genuinely tridiagonal at $t = 0$, dense for $t > 0$, and isotropic ($Q_t \rightarrow I$) as $t \rightarrow \infty$.

All quantitative statements are checked independently in `code/numerical_audit.py` (59/59 pass) and reproduced by the companion notebook.

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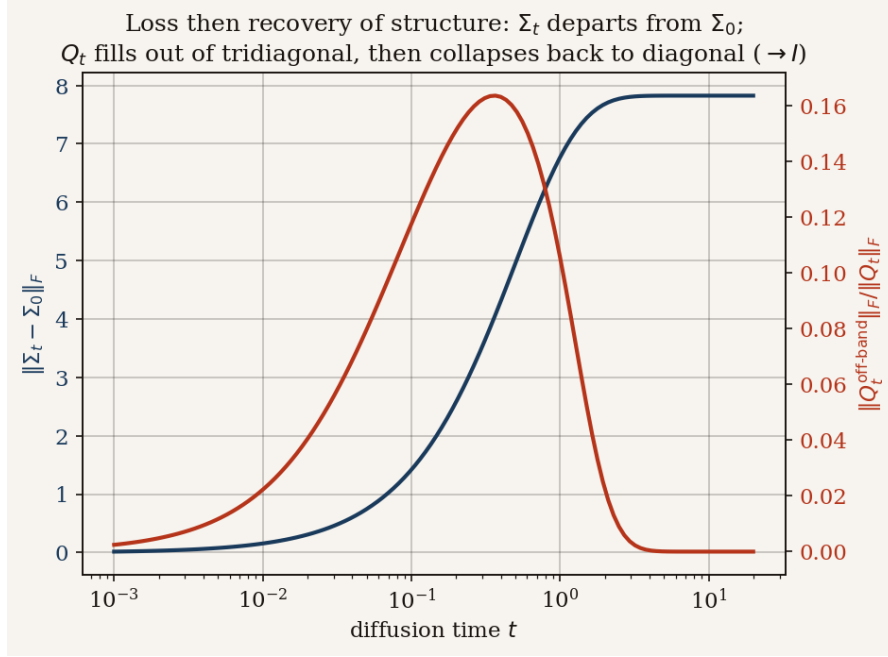


Figure 8: **Loss then recovery of structure.** $\|\Sigma_t - \Sigma_0\|_F$ (navy) grows monotonically as the covariance is driven to I ; the off-tridiagonal mass of Q_t (red) rises, peaks, and decays—the precision leaves and then returns to a diagonal form.

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